

N5 MAX: the final verdict

Five NVMe slots (four at PCIe Gen4 x1, one at Gen4 x4), a Ryzen AI Max+ 395 with 128 GB unified memory, and a seven-stage sealed campaign across ZFS, btrfs and mdadm, local LLM inference, a one-hour soak, USB ingest and drive-failure rebuilds. All figures come from sealed `result.json` files or the lab's own report tooling, as of 2026-10-08.

Verdict. This is an AI box first and a fast NAS second. Make the pool **ZFS RAIDZ1 5-wide**: 14.11 TiB, 7.83 GB/s cold reads, 93.51 % of reads kept with a drive out, a 29-minute rebuild. btrfs raid5 and mdadm raid5 read faster when healthy (9.034 / 9.035 GB/s), but btrfs raid5 falls to 5.87 % degraded and still carries the upstream write hole. Inference speed does not depend on the pool once a model is resident. Expect roughly 0.7128–0.9715× of Minisforum's published llama-bench floor.

7.83 GB/s
RAIDZ1 5-wide cold seq read
· 14.11 TiB

9.034 GB/s
btrfs raid5 cold seq read
(fastest, tied with mdadm
raid5 9.035)

93.51 %
RAIDZ1 seq read kept with
one drive out (btrfs raid5:
5.87 %)

×0.8976
gpt-oss-120b tg128 vs
vendor floor (53.23 vs 59.3
tok/s, HIP)

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1 · Executive summary

Each headline finding below carries its evidence: stage, layout, test id and the run timestamp of the sealed case.

- 1 Recommended default stays RAIDZ1 5-wide, and stage 7 now backs it.**

Degraded with one member out, RAIDZ1 still read 93.51 % of healthy sequential throughput and resilvered in 1,740 s at 1.1656 GB/s. btrfs raid5 fell to 5.87 % and took 3,720 s at 0.5498 GB/s.

`S7 · zfs-raidz1 / btrfs-raid5 · ST0-09 ST0-01_seq_read_1M_j4 · runs 2026-09-24-19.58 / 2026-09-24-05.07`
- 2 Fastest measured is now a tie: btrfs raid5 and mdadm raid5 (cold seq read).**

btrfs raid5 9.034 GB/s at 14.43 TiB, mdadm+ext4 raid5 9.035 GB/s at 14.41 TiB. Both are on the capacity × cold-read Pareto frontier from the lab's Pareto tool. RAIDZ1 5-wide reads 7.83 GB/s (1.154× behind btrfs raid5).

`S5 · btrfs-raid5 / mdadm-ext4-raid5 · ST0-01_seq_read_1M_j4 · runs 2026-09-14-13.57 / 2026-09-25-18.53`

3 The “mdadm is 8× slower” anomaly is resolved: it was a link fault, not mdadm.

SN1 had trained at Gen1 2.5 GT/s x1 in every original mdadm cell (#72), which pinned seq read at 1.14 GB/s. The reruns at Gen4 read 9.035 GB/s.

```
S5 · mdadm-ext4-raid5 · ST0-01_seq_read_1M_j4 · run 2026-09-25-18.53 (superseded 2026-09-15-00.41)
```

4 mdadm parity writes fast in sequence but collapses on 16K random writes.

mdadm raid5 has the highest cold sequential write (5.04 GB/s vs btrfs raid5 4.066), but only 2,489 IOPS on 16K random writes vs btrfs raid5 22,981. That is partial-stripe read-modify-write. mdadm raid6 is the same: 2,313 IOPS.

```
S5 · mdadm-ext4-raid5 · ST0-01_seq_write_1M_j4, ST0-02_rand_write_16k_j4 · run 2026-09-25-18.53
```

5 The ZFS-vs-btrfs write gap in the FS controls is a drive-state effect, not a filesystem verdict.

The ZFS stage-1 cells fully preconditioned their members before measuring, and the btrfs fs-control cells did not (per the lab’s backend notes). There btrfs raid5 wrote 4.066 GB/s vs RAIDZ1 0.718. Under ST0-09 S1 (same protocol, file-on-FS, both after the same 50 % incompressible fill) RAIDZ1 wrote 4.499 GB/s and btrfs raid5 2.78 GB/s.

```
S7 S1 · zfs-raidz1 / btrfs-raid5 · ST0-01_seq_write_1M_j4 · runs 2026-09-24-19.58 / 2026-09-24-05.07
```

6 The four x1 slots tax sequential work about 3×, but barely touch random and sync work.

One NM790 at x4 vs x1: seq read 5.888 vs 1.8 GB/s (3.27×), 4K random read 41,318 vs 35,116 IOPS (1.18×), sync 16K 1,398 vs 1,283 (1.09×). One x1 drive reaches 91 % of the 1.969 GB/s lane.

```
S1 · single-x1 / single-x4 · ST0-01, ST0-02, ST0-04 · runs 2026-09-13-13.00 / 2026-09-12-19.42
```

7 ZFS RAID10 is the sync-write layout; RAIDZ1 pays for parity on sync.

Cold sync 16K: RAID10 1,675 IOPS vs RAIDZ1 415. primarycache=metadata does not help sync (RAIDZ1 256), but it lifts 4K random reads by 50.2 % on RAIDZ1 and 58.7 % on RAID10.

```
S1/S2 · raid10 / raidz1 · ST0-04_sync_write_16k_j4, ST0-02_rand_read_4k_j4 · runs 2026-09-13-15.49 / 2026-09-13-17.14
```

8 Once a model is resident, the pool does not matter for inference.

Gemma 4 26B tg128 is 50.15 tok/s on btrfs and 50.08 on ZFS. Qwen3.8-27B (gufo-model-bench) decodes 12.44 tok/s from btrfs cold and 12.34 from ext4 warm. Pools differ only in cold load (5.035 s vs 6.758 s for Gemma).

```
S3 · btrfs raid5 / ZFS RAIDZ1 · AI-01 card + load rail · runs 2026-09-22-06.02 / 2026-09-23-04.57
```

9 We sit below the Minisforum vendor floor on all 16 comparisons.

Ours ÷ floor ranges from 0.7128 to 0.9715. gpt-oss-120b HIP is 0.8562 (pp512) and 0.8976 (tg128). Decode on Gemma 4 and Qwen3.5-122B is the furthest behind (0.7128, 0.7252 HIP). The vendor states no conditions beyond ngl 99 and FA on.

```
S3 · btrfs raid5 · AI-01 vendor group · run 2026-09-23-15.03
```

10 Vulkan decodes faster, HIP prefills faster, and Vulkan loads much slower.

Across the four floor models, Vulkan tg128 is 1.062–1.107× HIP and HIP pp512 is 1.038–1.082× Vulkan. Vulkan cold load takes 1.34–5.22× as long (gpt-oss-120b 49.375 s vs 11.256 s).

```
S3 · btrfs raid5 · AI-01 vendor group + load rail · HIP/Vulkan cells
```

11 Streaming models larger than memory is a load-time story, and ZFS loses it.

Kimi K2 (streamed) cold load took 38.334 s on btrfs vs 181.147 s on ZFS (primarycache=all). Decode is unusable either way: 1.09 / 1.55 tok/s.

S3 · btrfs raid5 / ZFS RAIDZ1 · AI-01 kimi-k2 streamed · runs 2026-09-29-03.21 / 2026-09-29-13.35

12 Thermals are a non-issue for an hour of sustained work.

In the 1-hour RAIDZ1 soak, mixed 70/30 held 0.886 GB/s and was steady, with drives peaking at 59.85 °C and the SoC at 52.0 °C. The sequential-write soak (1.178 GB/s) was not steady.

S4 · raidz1 · SOAK-02_mixed_70r30w_16k_j4 · run 2026-09-12-14.31

STANDING POSITIONS: WHERE THEY MOVED

Fastest measured was btrfs raid5 alone; it is now a tie with mdadm raid5. The mdadm reruns removed the link fault that hid it. **Recommended default** stays RAIDZ1, now with degraded and rebuild evidence behind it. **"mdadm 8× slower"** is withdrawn: #72 traced it to SN1 at Gen1; #55 is effectively answered. **btrfs raid5/6 write hole**: unchanged. #59 was never run (stage 8 deferred), so the upstream caveat stands as is.

2 · Setup & method

Platform

Minisforum N5 MAX (Ryzen AI Max+ 395, Radeon 8060S iGPU, 128 GB LPDDR5X unified memory). Five 4 TB Lexar NM790 NVMe drives, labelled SN1–SN5. SN1–SN4 sit on Gen4 x1 links (1.969 GB/s each) and SN5 on Gen4 x4 (7.88 GB/s). The NM790 is DRAM-less (HMB) and has no power-loss protection. Five 3.5" SATA bays hang off a JMB58x bridge on a shared Gen3 x2 uplink; they were **not** benchmarked. The box has 2×10GbE.

Campaign

Stage 1/2: ZFS layouts in cold (primarycache=none) and metadata regimes. Stage 3: 21 AI cells. Stage 4: one-hour soaks. Stage 5: btrfs and mdadm+ext4 controls, cold and warm. Stage 6: USB. Stage 7: drive failure and rebuild (STO-09). Stage 8: write-hole, deferred.

Rules applied in this report

Storage claims use **Protocol B** cases (derived durations; `params.protocol` starting with `B-`) and the lab selector's **published** case per subject (the lab's publish selector). Non-steady cases (`rails.steadiness.steady == false`) are drawn hatched or flagged red. PROVISIONAL ($\geq 2\times$ spread against the subject's own history) and RUN SUPERSEDED figures are marked and never carried into a conclusion. `deprecated-fixed60`, `preliminary-omega`, calibrate-mode cells and link-downtrained runs are excluded.

Regimes. *Cold* (ZFS primarycache=none, or dropped caches) is the layout story. *Metadata* and *warm* are separate regimes. Warm sequential reads converge on the DRAM ceiling (35.685–36.74 GB/s whatever the layout), so they are not layout evidence.

Access method. ZFS stage 1/2 random, mixed and sync tests ran on zvols; btrfs and mdadm ran on files. Random IO is compared within a stack only. Sequential STO-01 is file-on-FS everywhere.

Selector accounting at build time: 252 run directories, 964 published subjects. Case verdicts: VOID 250, PASS 1428, FAIL 127, UNMEASURED 37. Coverage of the campaign matrix:

Campaign matrix cells and whether a sealed result.json backs them. Stage-7 "failed" = arms whose S4 write return failed the harness rule, not crashed cells.

stage	sealed	no result.json	lost	matrix status
Stage 1	13	0	0	passed 13
Stage 2	10	0	0	passed 10
Stage 3	21	0	0	passed 21
Stage 4	2	0	0	passed 2

stage	sealed	no result.json	lost	matrix status
Stage 5	12	0	0	passed 12
Stage 6	0	0	1	passed 1
Stage 7	10	0	0	passed 3, failed 7
Stage 8	0	0	0	deferred (write-hole, #59)

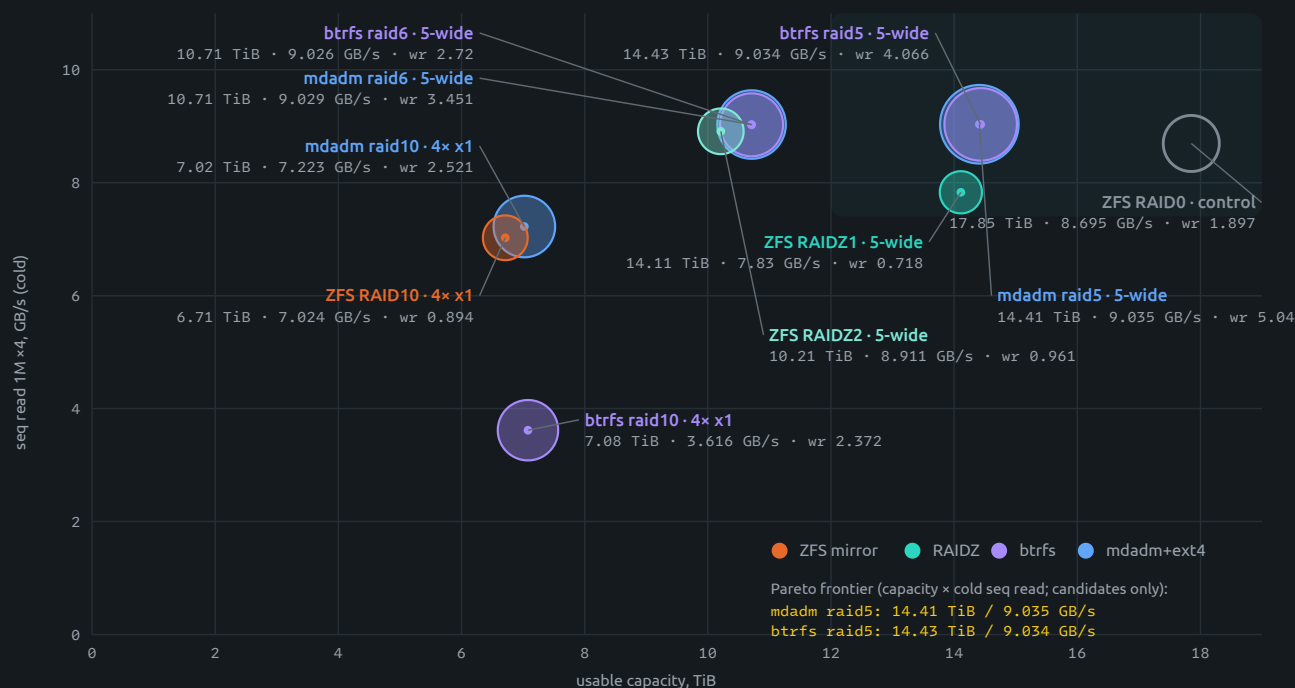
3 · Storage findings

3.1 Layouts: the tradeoff compass

Plotting capacity against cold sequential read shows the whole decision on one chart. The parity layouts on all five drives cluster at the top. With SN5 at x4 they read 9.0 GB/s (btrfs raid5/6, mdadm raid5/6) and 8.911 GB/s (RAIDZ2). RAIDZ1 sits a step lower at 7.83 GB/s with the second-highest usable capacity of any candidate (14.11 TiB, against 14.43 TiB for btrfs raid5). The mirrors pay half the raw capacity: ZFS RAID10 holds 6.71 TiB at 7.024 GB/s.

Tradeoff compass: capacity × cold sequential read

Cold regime (primarycache=none / dropped caches), Protocol B published cases. Bubble area = cold seq write 1M×4.



Filled = steady on both axes; ring = control layout, not a candidate. ZFS cells were fully preconditioned, btrfs/mdadm cells not: seq write differs by drive state (see STO-09 S1).

Capacity × cold sequential read; bubble area is cold sequential write. Pareto frontier computed the same way as the lab's Pareto tool. RAID0 is a control.

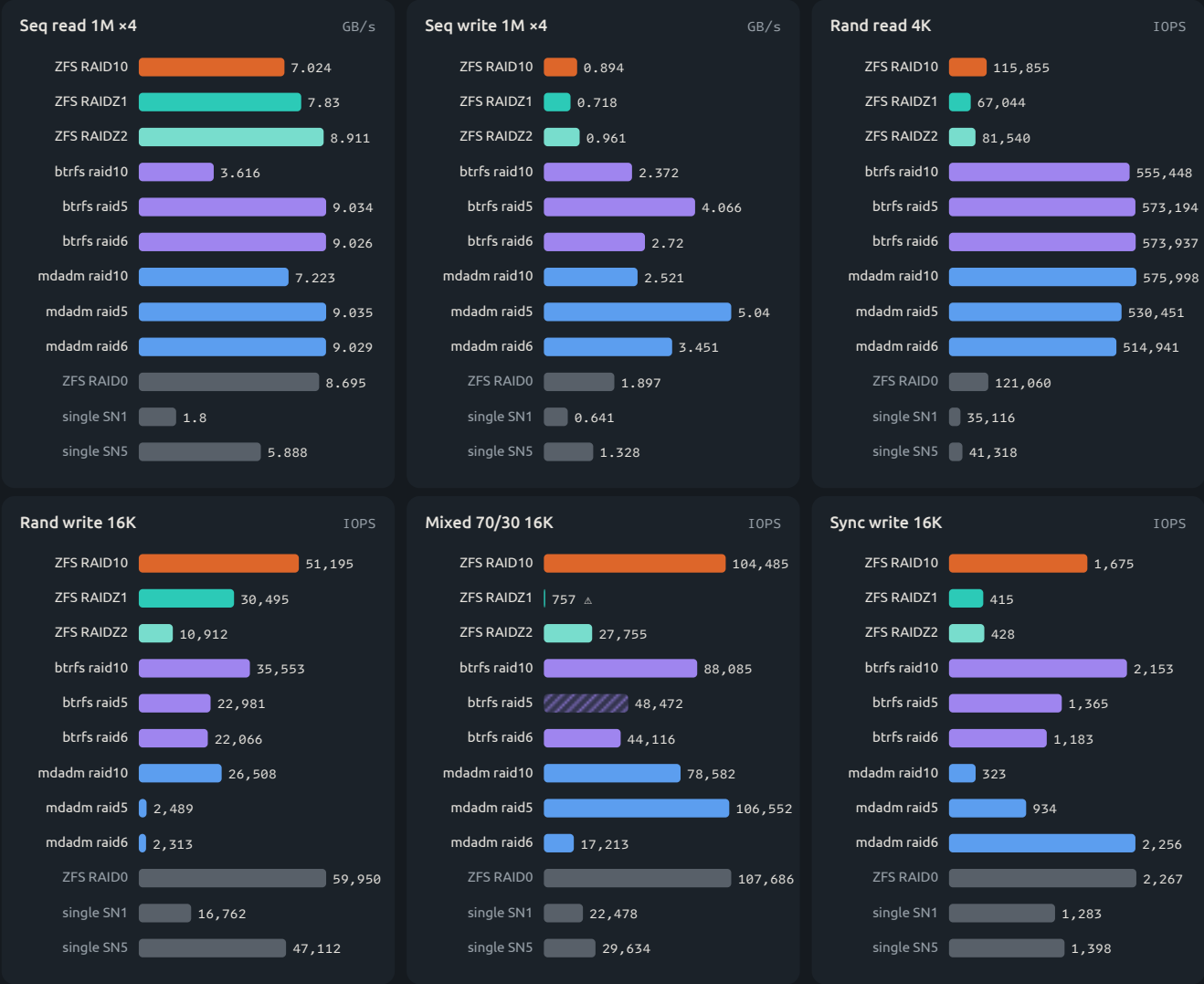
ANOMALY KEPT ON THE RECORD

btrfs raid10 reads only 3.616 GB/s cold, against 7.223 for mdadm raid10 on the same four x1 drives. Two ramp replicates (30 s and 210 s) reproduce it. The likely cause is btrfs raid10 read-mirror selection (PID-based, so a single reader stays on one mirror). That is unproven here; see gaps.

3.2 Every layout, six workloads

Small multiples: every cold layout, six workloads

Protocol B published cases, cold regime. Hatched = non-steady (rails.steadiness). Δ = PROVISIONAL (≥2× spread vs own history).



ZFS random/mixed/sync ran on zvols, btrfs and mdadm on files: compare random IO within a stack, not across. Seq read/write (STO-01) is file-on-FS everywhere.

Small multiples of the cold published cases. Within a stack, read across a row; across stacks, compare only the seq-read and seq-write panels (file-on-FS everywhere).

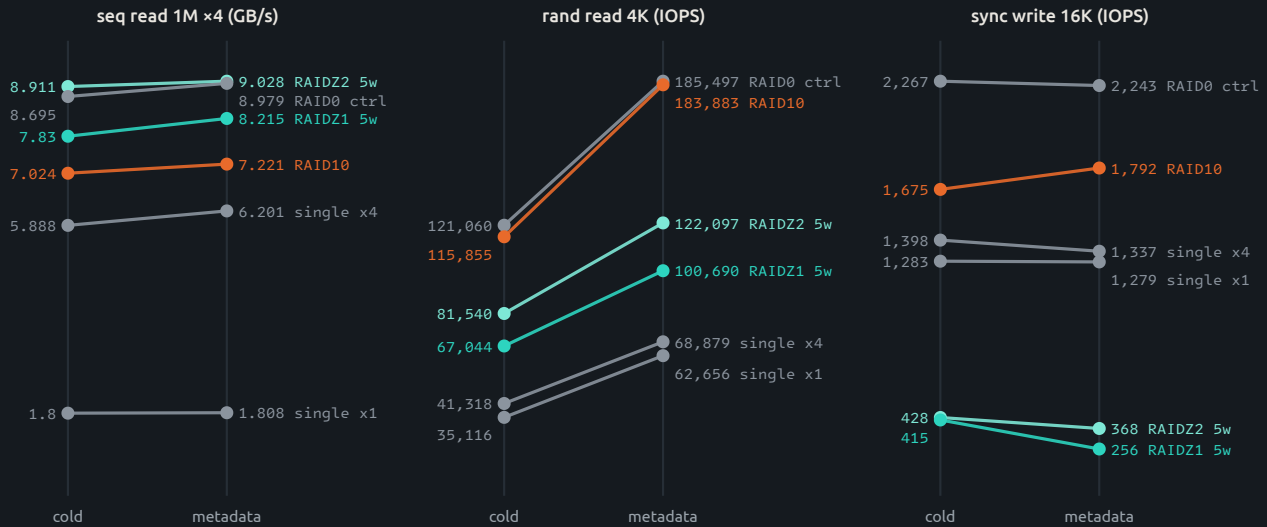
Three patterns hold up. **(1)** Every parity layout reaches 9.0 GB/s sequential reads once SN5's x4 lane is a member. The x1 members are the limit, not the filesystem. **(2)** Sync writes favour mirrors on ZFS (RAID10 1,675 vs RAIDZ1 415 IOPS). On files they favour btrfs over mdadm raid10 (2,153 vs 323). **(3)** mdadm parity is a sequential-write specialist: highest seq write (5.04 GB/s), worst 16K random write (2,489 IOPS).

RAIDZ1 cold mixed 70/30 (757 IOPS) is **PROVISIONAL**: its own prior good value was 14,582 IOPS, a 19× spread (lab storage report). Do not quote it.

3.3 Cold vs metadata

Cold vs metadata: the cache regime changes the story

ZFS primarycache=none → primarycache=metadata, same pool, same members. Random 4K reads lift; sync writes do not.



Hollow end = non-steady by rails.steadiness. Random/sync on ZFS run on zvols. Source: Protocol B published cases.

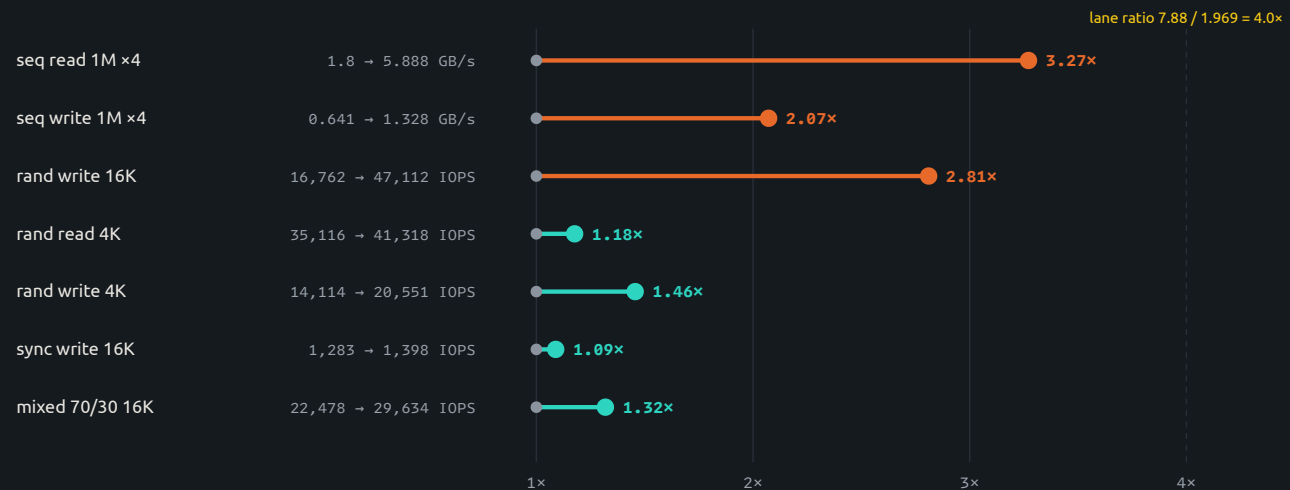
Same pool, same members; only the ARC policy changes.

primarycache=metadata keeps the indirect blocks in ARC, so random reads stop paying for metadata fetches. 4K random read rises 50.2 % on RAIDZ1 and 58.7 % on RAID10. Sync writes do not improve, and on RAIDZ1 they fall (415 → 256 IOPS). Sequential reads barely move. Production pools would normally run primarycache=all, which is warmer still, so the cold numbers here are a **floor**.

3.4 The x1 lane tax

The x1 lane tax: one NM790 on Gen4 x1 vs Gen4 x4

Same drive model, ZFS single-drive pool, cold. Ratio x4 ÷ x1 per test; the dashed line is the 4.0× lane-bandwidth ratio.



Sequential work pays the lane tax; small-block random and sync work is drive-bound, so the four x1 slots cost little there.

Single-drive pools, cold. The lane ratio is 4.0×; only sequential work comes near it.

One x1 drive reads 1.8 GB/s, 91 % of its 1.969 GB/s lane, and ZFS RAID10 over four x1 drives reads 3.90× that. So the striping works, and the lanes are the ceiling. Small-block random and sync IO are bound by the drive's controller and flash, not the link. The four x1 slots are therefore a sequential-bandwidth tax, not a responsiveness tax.

3.5 ZFS vs btrfs vs mdadm

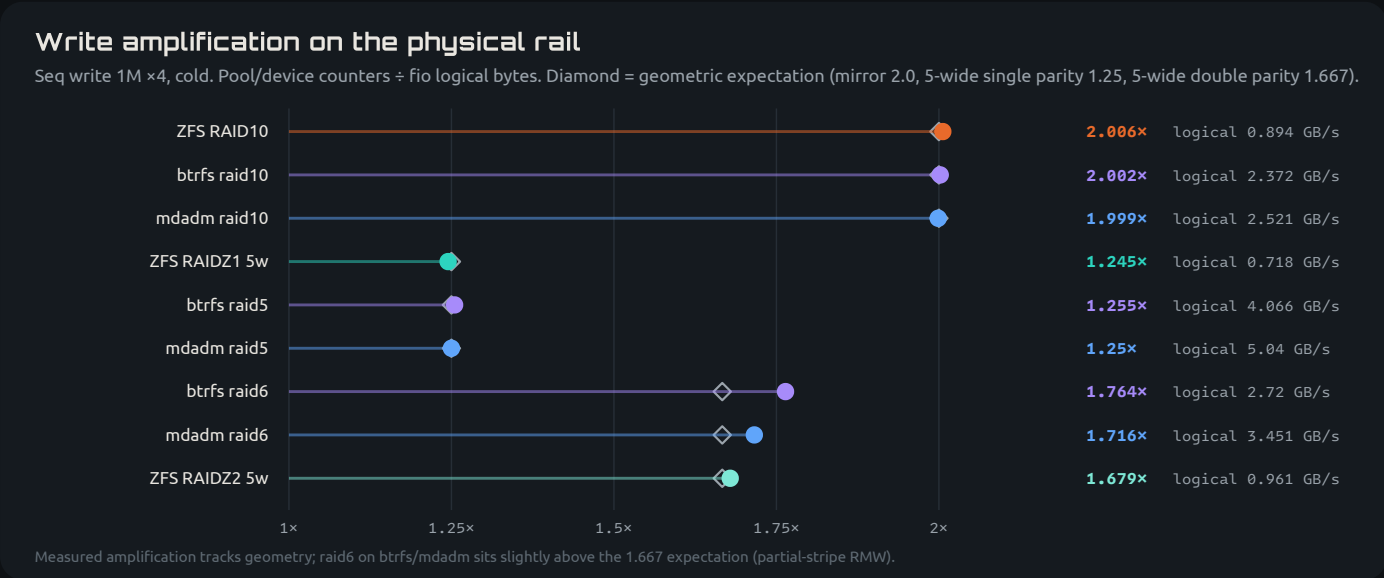
Healthy sequential. On the fs-control battery, btrfs raid5 out-reads RAIDZ1 (9.034 vs 7.83 GB/s at j4; 9.033 vs 6.748 at j1) and out-writes it by 5.66× (4.066 vs 0.718). **That write gap does not survive a like-for-like test.** STO-09 S1 runs the same file-on-FS protocol on both

after the same 50 % incompressible fill. There RAIDZ1 wrote **4.499 GB/s** and btrfs raid5 **2.78**. Reads kept their order: RAIDZ1 7.434 vs btrfs 9.045 GB/s.

The likeliest explanation is NM790 write state. The ZFS stage-1 cells fully wrote their zvols and benchmark file before measuring. The btrfs fs-control cells got no equivalent preconditioning (mkfs.btrfs takes seconds), as the lab notes in its backend code. The soak and stage-7 control show this drive slows sharply once its SLC cache is spent (§3.7, §6). The fs-control seq-write gap therefore compares different drive states, not filesystems.

Random IO cannot be ranked across stacks from this campaign: ZFS used zvols (volblocksize-bound), btrfs and mdadm used files. In STO-09 S1, ZFS 4K random reads on a recordsize=1M dataset ran at 0.033 GB/s against btrfs 2.226 GB/s. That is mostly whole-record read amplification of 1M records. It is a tuning artefact, and the reason a VM or database dataset needs a small recordsize or a zvol.

3.6 Write amplification



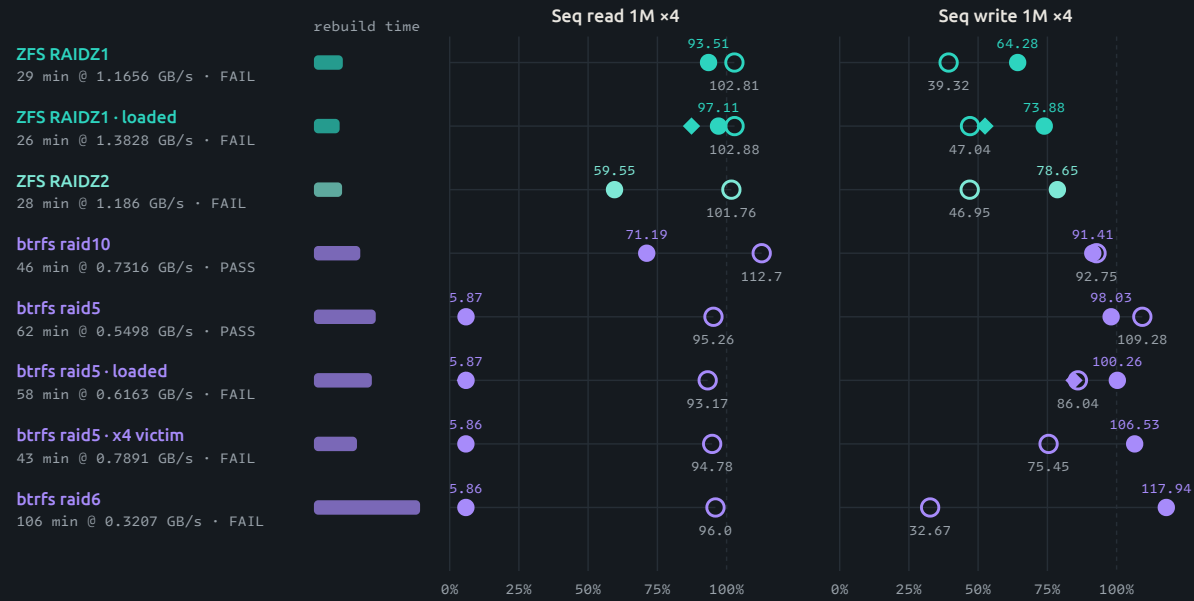
Physical rail (pool or device counters) ÷ fio logical bytes for seq write 1M ×4.

Every layout amplifies writes by what its geometry predicts: mirrors 2.006x (ZFS), 2.002x (btrfs), 1.999x (mdadm); 5-wide single parity 1.245x / 1.255x / 1.25x. Double parity sits slightly above the 5/3 expectation on btrfs (1.764x) and mdadm (1.716x), and close to it on RAIDZ2 (1.679x). For endurance on a no-PLP, DRAM-less drive, RAIDZ1 writes 38 % fewer flash bytes per logical byte than a ZFS mirror.

3.7 Degraded, rebuild and the write return (stage 7)

Fail one drive: degraded throughput, rebuild, and the return

STO-09, 50 % fill, file on FS. ● degraded idle S2 · ◆ rebuilding S3 (% of healthy ref = better of S1/S4) · ○ healthy-again S4 (% of S1; FAIL = >10 % below).



Every arm's reads came back (S4 seq read 93.17–112.7 %). The S4 seq-write shortfall is partly drive write-state: a no-failure control returned 66.58 % / 67.71 % of S1 (n=2).

STO-09 per arm. The control rows are in the appendix table. Loaded arms ran fio during the rebuild.

Reads with a drive out. ZFS reconstructs from parity at nearly full speed: RAIDZ1 93.51 %, loaded 97.11 %, RAIDZ2 59.55 %. btrfs raid5 and raid6 collapse to about 0.53 GB/s (5.87 % / 5.86 %), whichever member failed, including the x4 SN5 (5.86 %). btrfs raid10 keeps 71.19 %.

Rebuild. ZFS resilvers only allocated blocks, at 1.1656–1.3828 GB/s (29–28 min at 50 % fill). btrfs replace ran 0.5498 GB/s on raid5 (62 min) and 0.3207 GB/s on raid6 (106 min). The replacement received its allocated-only share as predicted (observed/predicted 1.002 on zfs-raidz1, lab storage report).

The write return. All arms got their reads back (S4 93.17–112.7 % of S1). Every ZFS arm and btrfs raid6 failed the harness's S4 write test (more than 10 % below S1): RAIDZ1 39.32 %, RAIDZ2 46.95 %, btrfs raid6 32.67 %.

The preregistered no-failure control ran the same sequence with no failure and no rebuild. It returned 66.58 % and 67.71 % of S1, with S4/S2 = 0.9986 / 1.0167. So roughly two-thirds of the S4 level is what these drives deliver after the write volume of S1–S2 alone. The *extra* loss in the failure arms (RAIDZ1 down to 39.32 %) is failure- or rebuild-specific by the preregistered rule.

The lab report places the knee at about 2.3–2.4 TB written per survivor. Idle and TRIM recovery rungs did not restore the rate. This is a drive write-state and endurance characteristic; it says nothing about data integrity either way.

BTRFS RAID5/6 DATA WRITE HOLE

Unmitigated upstream. `-m raid1c3` protects metadata only. The lab's write-hole harness (#59) was built but never run (stage 8 deferred; a sysrq blocker), so this campaign adds no data either way. Combined with the 5.87 % degraded read, btrfs raid5/6 is not recommended for data you cannot re-download.

3.8 Soak and thermals (stage 4)

One-hour soaks on RAIDZ1 and RAID0, metadata regime. RAIDZ1 mixed 70/30 held 0.886 GB/s (54,107 IOPS, steady). Drives peaked at 59.85 °C and the SoC at 52.0 °C. RAID0 mixed held 1.616 GB/s (steady, drives 60.85 °C). The sequential-write soaks were **not steady**: RAIDZ1 1.178 GB/s, RAID0 1.473 GB/s. That is the same post-cache write sag as in stage 7, not thermal throttling: drive temperatures stayed under 60 °C.

3.9 USB ingest (stage 6)

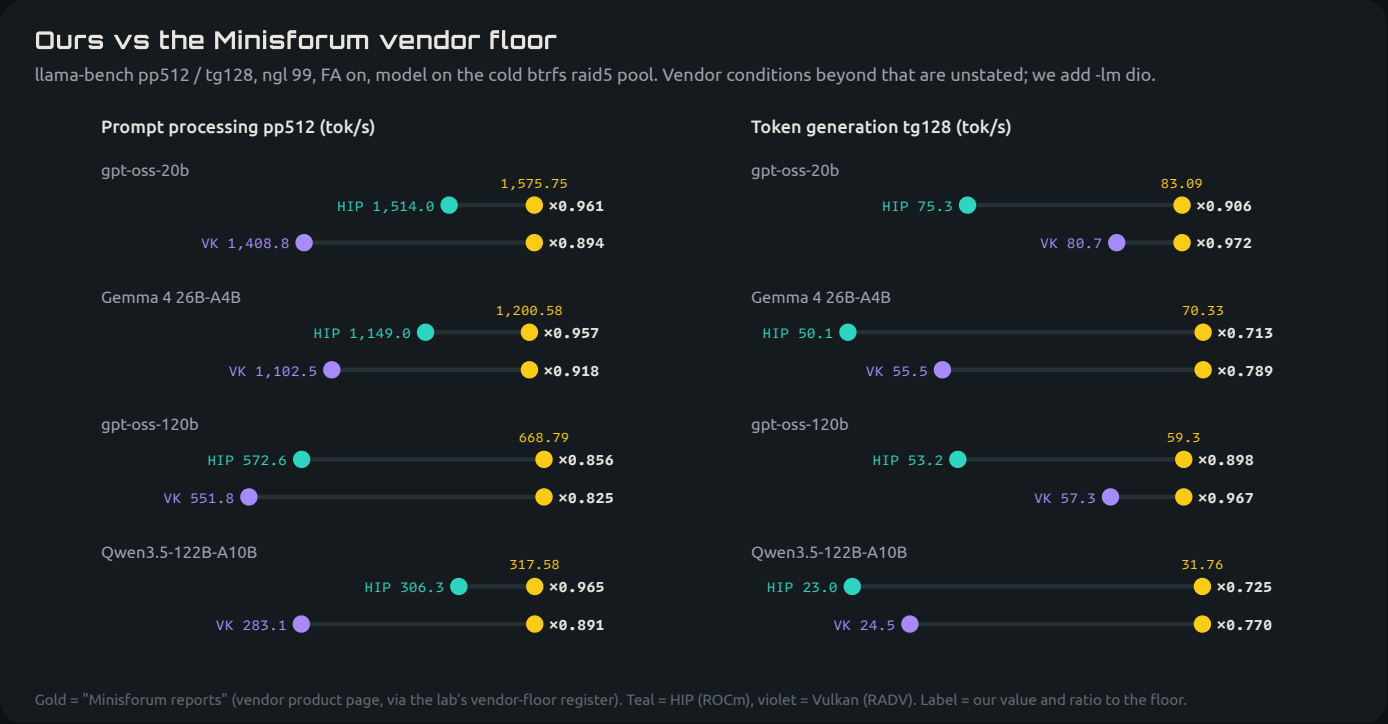
The campaign’s stage-6 STO-13 cell passed, but its result was **lost**: it was written to a tmpfs scratchpad that a host reboot wiped. The post-campaign variant STO-13b (a USB-attached NVMe ↔ RAIDZ1 5-wide, 200.74 GiB corpus, different OS image) recorded **905.6 MB/s** reading from USB into the pool and **376.2 MB/s** writing out (rsync including the sync drain). The pool was not the limit in either direction. The enclosure fell back from UAS to usb-storage, so treat these as a floor for this enclosure, not for the port. Integrity: 0 failures.

4 · AI findings

Stage 3 ran llama.cpp (HIP/ROCm and Vulkan/RADV) on 11 models, from a 0.6B smoke model to Kimi K2 streamed from disk. Most cells load from cold btrfs raid5; paired cells load from ZFS RAIDZ1 (primarycache=all, ARC cold at start). All 21 cells passed. The lab’s status table still lists 21 per-test rows as UNMEASURED (#67/#68). Full table in the appendix.

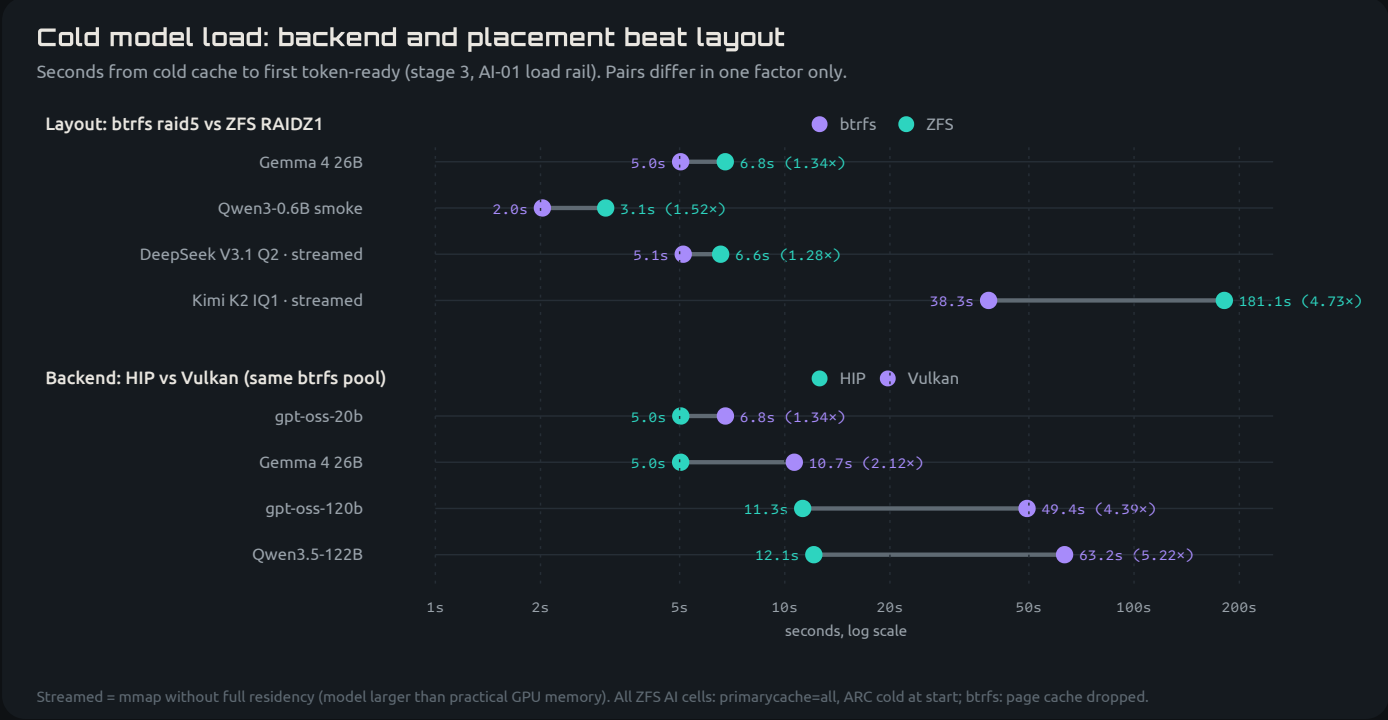
4.1 Throughput per model

Resident models (HIP, pp512 / tg128 tok/s): gpt-oss-20b 1,488.02 / 75.49; Gemma 4 26B-A4B 1,130.87 / 50.15; gpt-oss-120b 564.01 / 53.12; Qwen3.5-122B-A10B 308.21 / 22.97; dense Qwen3.8-27B Q4 332.94 / 12.59; DeepSeek V4 Flash IQ2_XXS 27.35 / 15.26. Streamed models are not interactive: DeepSeek V3.1 Q2 1.78 and Kimi K2 1.09 tok/s decode.



Ours vs the Minisforum floor for the four models it publishes. Values in the appendix vendor table.

4.2 Cold load, resident, streamed

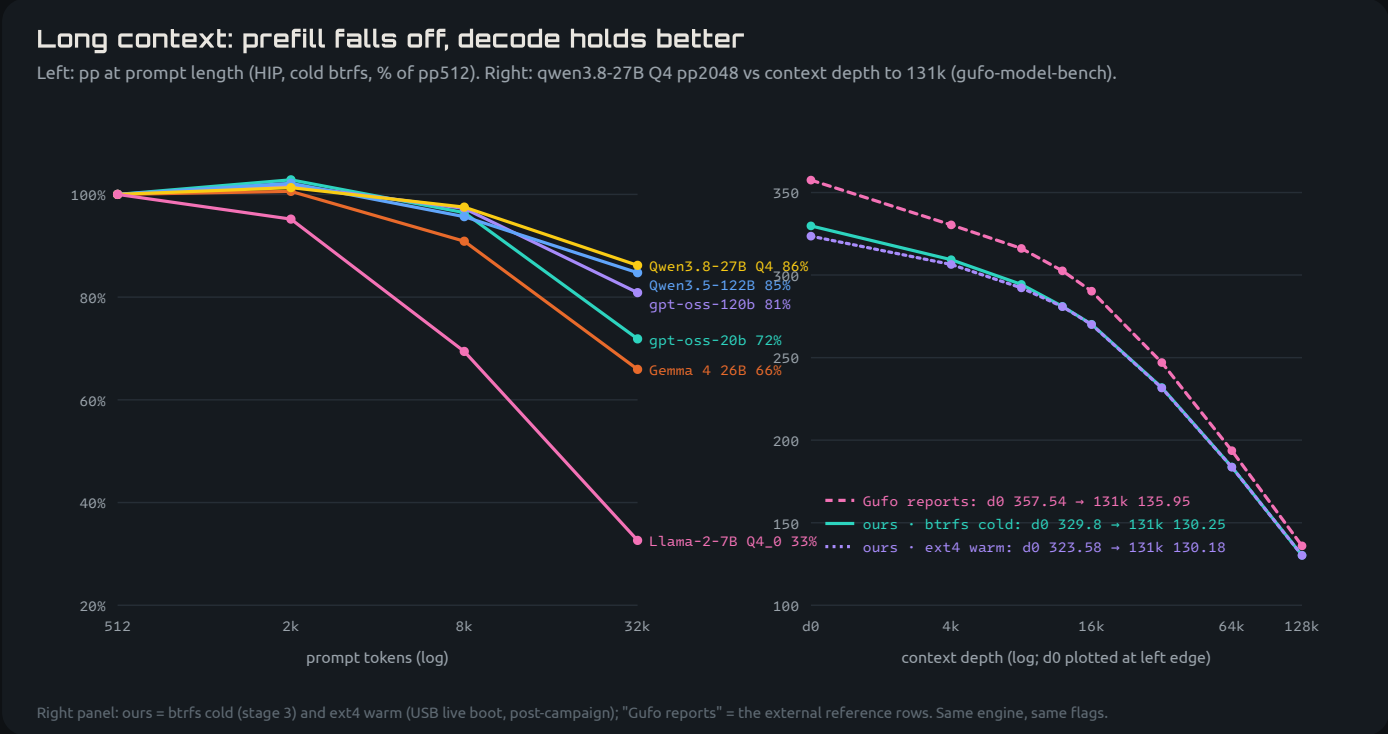


Load time pairs that differ in one factor. Log scale.

Resident. Layout is irrelevant to pp and tg. Gemma 4 on btrfs vs ZFS: 1,130.87 vs 1,136.82 pp512, 50.15 vs 50.08 tg128. Qwen3.8-27B decodes 12.44 tok/s off cold btrfs and 12.34 off a warm ext4 USB boot. **Cold load** is where pools differ, and only modestly for resident models: 5.035 s btrfs vs 6.758 s ZFS for Gemma. **Backend** matters more than layout: Vulkan cold loads take 1.34–5.22× as long as HIP on the same pool. **Streamed** models (larger than practical GPU memory, read via mmap with ~21× read amplification) load 4.73× slower from ZFS (Kimi K2), although decode stays at 1–2 tok/s either way.

The DeepSeek V3.1 streamed cell’s load-rate field (49.927 GB/s) is mmap-deferred bytes over wall time, not a pool rate. Its pool rail read 3.761 GB/s.

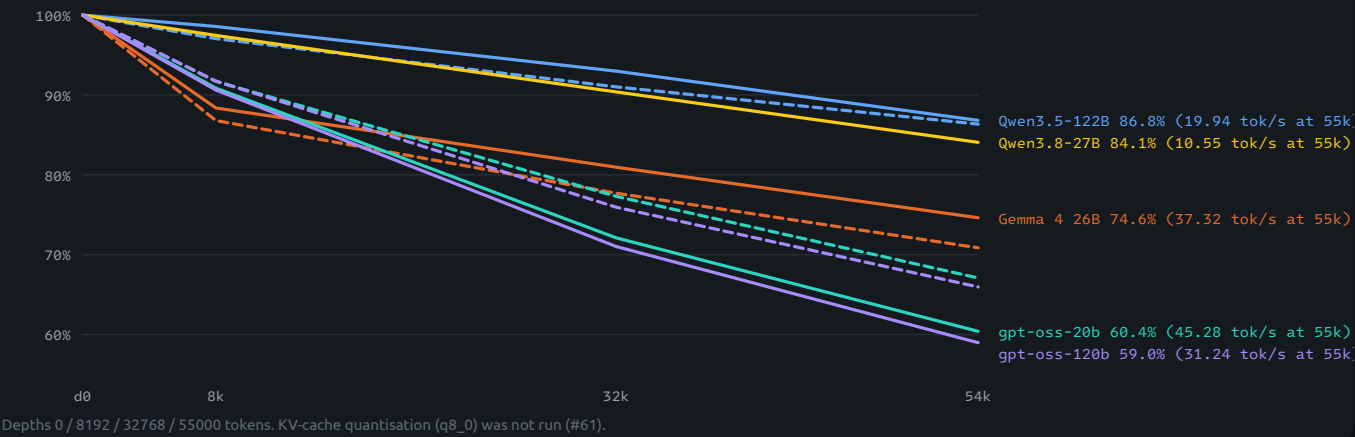
4.3 Prefill and decode with context



Left: stage-3 prefill curves to 32k. Right: gufo-model-bench depth sweep to 131k for Qwen3.8-27B Q4.

Decode retention with context depth (tg128 at depth ÷ depth 0)

Stage 3, cold btrfs, f16 KV cache. Solid = HIP, dashed = Vulkan. The fastest decoders lose the largest share as KV reads grow.



Decode retention to 55k tokens, f16 KV.

Prefill holds within a few percent to 8k for the MoE models and drops to 457.82 tok/s at 32k for gpt-oss-120b (81 % of pp512). The dense Llama-2-7B falls hardest (432.08 tok/s at 32k, 33 %).

Decode at 55k keeps 86.8 % on Qwen3.5-122B but only 59.0 % on gpt-oss-120b (31.24 tok/s). The fastest decoders lose the largest share, because the per-token KV read is a bigger fraction of a fast token.

Stage-3 curves stop at 32k (prefill) and 55k (decode). The gufo-model-bench rows carry Qwen3.8-27B to 131,072 tokens: pp2048 329.8 → 130.25 tok/s.

KV-cache quantisation was not run. Every cell used f16 KV. There are no q8_0 rows (#61).

Speculative decoding (USB live boot, ext4, post-campaign, gufo DFlash2): Qwen3.8-27B tg128 rises from 12.34 to 22.68 tok/s on prose (1.84×). That is the largest single decode lever measured on this box for a dense model.

4.4 Which layout helps AI

For resident models, none: put models wherever capacity is convenient. Cold load favours btrfs slightly for resident models (5.035 vs 6.758 s) and strongly for streamed ones. If you plan to stream models larger than memory, keep them on a non-ARC filesystem or a dataset with primarycache=metadata. The second option is a recommendation from mechanism; it was not tested here. For a RAG box the deciding properties are the storage ones: degraded reads, rebuild time, and capacity for embeddings and documents. All three point to RAIDZ1.

5 • Competitive analysis

External figures are quoted as “X reports Y”, with condition differences noted. None of them is our result.

Source	Model / test	They report	Ours (sealed)	Conditions differ
Minisforum (vendor floor)	gpt-oss-120b pp512/tg128	668.79 / 59.3	572.64 / 53.23	llama-bench ngl 99 FA on; nothing else stated. We add -lm dio.
Minisforum (vendor floor)	Qwen3.5-122B / gpt-oss-20b / Gemma 4 26B tg128	31.76 / 83.09 / 70.33	23.03 / 75.31 / 50.13 (HIP)	as above. Decode is our largest shortfall.
ITPro	gpt-oss-20b decode, Vulkan	72.9 tok/s (ROCm “40+”)	80.70 VK / 75.49 HIP	Ollama-style chat prompt vs llama-bench tg128; their ROCm stack evidently older.
ITPro	Gemma 4 26B decode	44 tok/s	50.15 HIP / 55.40 VK	Ollama, default ~30 GB GPU memory ceiling.
ServeTheHome	Beelink GTR9 Pro (same silicon) gpt-oss-120b	31.41 tok/s	53.12	LM Studio, out-of-box 96 GB allocation, different chassis.
ServeTheHome	N5 Max, Qwen3.6 27B dense	~9 tok/s	12.59 (Qwen3.8-27B Q4)	Different model generation and FP8 vs Q4 weights: shape only.

Source	Model / test	They report	Ours (sealed)	Conditions differ
MLPerf v6.1 edge (Atlas)	Qwen3.6-27B NVFP4, agentic replay	10.216 output tok/s of wall	12.59 tg128	Different metric (end-to-end wall), engine and model.
Lin / llm-tracker	Llama-2-7B Q4_0 Vulkan pp512/tg128	881.71 / 52.22	1,366.27 / 54.16 (HIP)	May 2025 stack, TheBloke conversion.
kyuz0 toolboxes	Llama-2-7B Q4_0 pp512	RAADV 1337.70 · ROCm 7.2.3 1545.36	1,366.27 (HIP)	Newer ROCm; we are 12 % behind their ROCm figure.
Gufo (gufo-model-bench)	Qwen3.8-27B Q4 pp2048 d0 / tg128 d0	357.54 / 12.06	329.8 / 12.44	Same engine and flags (matched rows). Ours pp is 92.2 % of theirs, tg 103.2 %.
Halogen	decode retention at 32k	gpt-oss-120b 92.6 %	70.8 % (compare-external)	Closed engine. Shape only; absolute numbers blocked.
Level1Techs / AMD	Llama 3.1 70B 4.97 tok/s · Llama 4 Scout “up to 15”	–	not measured	No matching model cell. Context only.

Reading. On llama-bench we land 3.5–4.3 % below the vendor’s prefill numbers for three of four models (HIP; gpt-oss-120b 14.4 %) and 9.4–28.7 % below its decode numbers. The vendor’s conditions are unstated, so this is a floor comparison and not a defect claim. Against reviewers using consumer front-ends (Ollama, LM Studio), our llama.cpp numbers are higher, as expected from the leaner engine. Against same-engine community references (Gufo), we match decode and trail prefill by about 8 %. Against newer ROCm toolboxes (kyuz0), we trail prefill by about 12 %. The gap tracks the software stack, not the box.

As a NAS. Every candidate pool reads well above what one 10GbE link carries (1.25 GB/s), so a single client will be network-bound on reads. In the post-cache write state, though, RAIDZ1 sequential write (0.718 GB/s cold, stage 1) is *below* 10GbE line rate. Sustained large ingest can therefore be drive-bound on this NM790 population. Network throughput itself was not measured in this campaign.

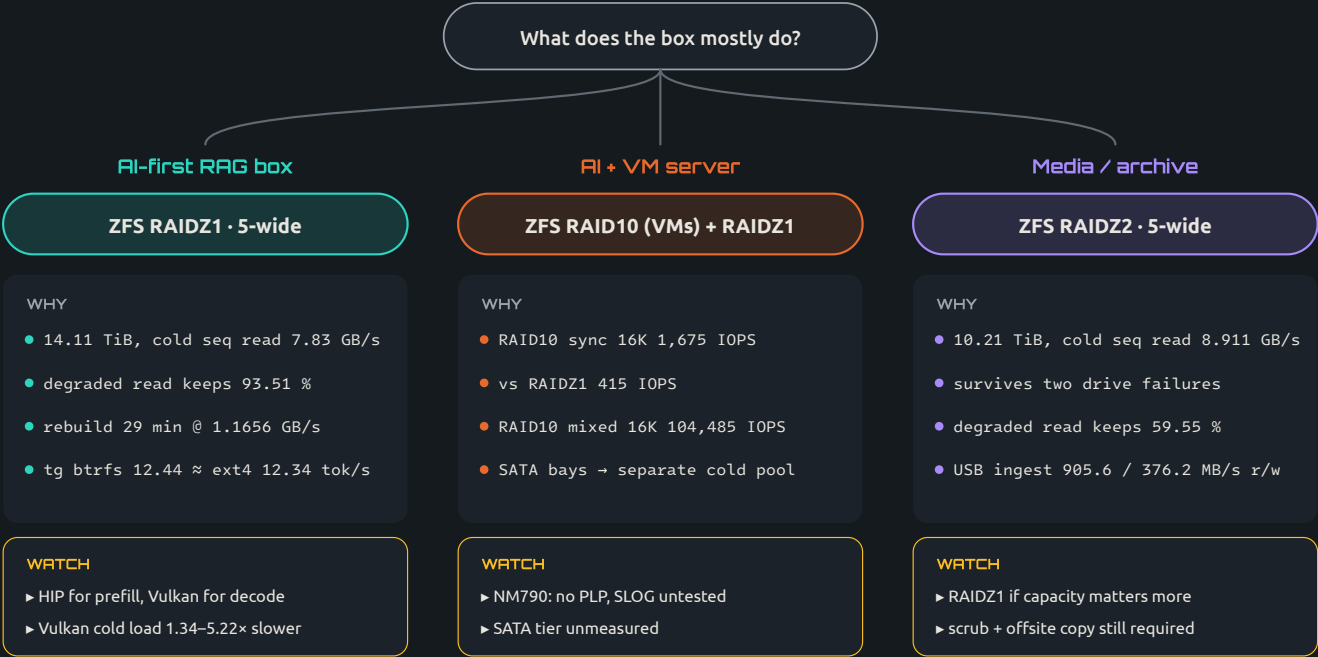
6 • Cross-test insights

- **Drive write state outranks the filesystem.** The same RAIDZ1 layout wrote 4.499 GB/s in STO-09 S1 (after a 50 % fill), 0.718 GB/s after full preconditioning (stage 1), and 1.178 GB/s (non-steady) in the one-hour soak. The no-failure control lost a third of S1 without any failure. Any write comparison on NM790s must state the drive state.
- **SN5 at x4 is what makes the 5-wide layouts fast.** Every 5-member parity layout reads 9.0 GB/s. The superseded 4-wide RAIDZ1 on four x1 members read 6.467 GB/s (RUN SUPERSEDED, comparison only).
- **Link training is a silent failure mode.** SN1 trained at Gen1 in six cells (#72) and again on 2026-09-29 (that AI run is excluded). A per-run link-speed gate is cheap insurance; see gaps.
- **Warm reads hide layout.** All six warm FS controls read 35.685–36.74 GB/s, the DRAM ceiling. Small-file copy (STO-06) converges at 0.418–0.724 GB/s across every cold layout. Neither separates layouts.
- **AI and storage barely interact once loaded.** Inference reads the model once. After that, the only storage-sensitive AI behaviour is cold load and streaming. Pick the pool on storage merits.

7 · Recommendations

Recipe: which pool for which box

A decision path built from the sealed evidence. Numbers are the published cold-regime cases quoted in the report.



Any workload: btrfs raid5/6 is the fastest healthy pool, but not for data you cannot re-download.
Degraded seq read 5.87 % of healthy · rebuild 62 min · write-hole unmitigated (raid1c3 = metadata only)

SATA bays (5×3.5", JMB58x Gen3 x2 shared) were not benchmarked; SLOG (#57) and the btrfs write-hole test (#59) were not run.

Decision graphic. Every number on it is pulled from the published cases.

Per workload

AI-FIRST RAG BOX

ZFS RAIDZ1 across all five NVMe (14.11 TiB). Models on a recordsize=1M dataset (as tested). Use a small-recordsize dataset or a zvol for the vector DB. HIP for batch prefill and embeddings, Vulkan for interactive decode (+6–11 % tg), and accept Vulkan’s slower cold load. Models larger than memory: keep them off ARC (primarycache=metadata) or on a btrfs volume. For dense models, try speculative decoding.

AI + VM SERVER

Two NVMe pools trade capacity for sync IOPS: RAID10 for VM disks (sync 1,675 IOPS, mixed 104,485) vs RAIDZ1 for everything else. Alternatively, one RAIDZ1 pool with VM zvols if capacity matters more. Use the five 3.5" SATA bays as a separate cold ZFS pool (RAIDZ1/RAIDZ2) for backups and media. It is **unmeasured** and shares a Gen3 x2 uplink, so do not put VMs on it. No PLP: do not set sync=disabled; a SLOG was not tested.

MEDIA / ARCHIVE

RAIDZ2 5-wide (10.21 TiB, cold read 8.911 GB/s, survives two failures, 59.55 % degraded read) or RAIDZ1 for capacity. Scrub monthly. Large ingest over USB runs about 0.9 GB/s on a UAS-less enclosure.

NOT RECOMMENDED

btrfs raid5/6 for primary data: write hole, 5.87 % degraded reads, slow replace. mdadm raid5/6 for VM or database work: 16K random writes collapse to 2,313–2,489 IOPS. btrfs raid10 for single-stream reads (3.616 GB/s).

Per OS / stack

Stack	Do	Avoid / note
Proxmox VE	ZFS RAIDZ1 (or RAID10 + RAIDZ1 split) for local-zfs. VM disks as zvols (16K volblocksize default; not A/B-tested here). Pass the iGPU to an LXC for llama.cpp, or run it on the host.	Check <code>lspci -vv</code> link speeds after every boot (SN1 downtrained twice). Separate pool for SATA bays.
TrueNAS SCALE	RAIDZ1 or RAIDZ2 pool. SMB/NFS datasets recordsize=1M for media and models, 16–64K for shares with small files.	AI apps run in containers; ROCm support on TrueNAS was not tested. No SLOG without PLP.
Fedora / Ubuntu	OpenZFS RAIDZ1 for data. Current ROCm or Mesa RADV for llama.cpp (newer stacks reach the community's higher pp). Keep models on ZFS or btrfs; resident speed is identical.	If you choose btrfs, use raid1/raid10 profiles for data (raid1c3 metadata). mdadm raid5 suits bulk sequential only.

8 · Buying guidance

- **Buy it** if you want one quiet box that serves files at 10GbE and runs 20–120B MoE models at interactive speed: gpt-oss-120b at 53.12 tok/s, Qwen3.5-122B at 22.97 tok/s.
- **Do not expect** NVMe-array sequential numbers beyond about 9 GB/s, or more than about 2 GB/s from any one of the four x1 slots. Small-block IO is unaffected.
- **Drives:** the NM790 is DRAM-less with no PLP, and its sustained write falls sharply after the SLC cache (\$3.7). For write-heavy or sync-heavy use, prefer drives with DRAM and PLP, especially in the x4 slot.
- **Dense models** are bandwidth-bound: ours decodes Qwen3.8-27B Q4 at 12.59 tok/s; ServeTheHome reports ~9 tok/s for Qwen3.6 27B (FP8) and Level1Techs reports 4.97 tok/s for Llama 3.1 70B on this silicon. MoE models are where the box shines.
- **Expect software to move results** more than storage does: ROCm version, backend and engine shift pp by 10–25 %.

9 · Limitations & open questions

- Write-hole test (#59) not run; SLOG (#57) not run; KV-cache quantisation (#61) not run; prefill past 32k in stage 3 (#60) covered only by the Gufo rows to 131k; scheduling probe (#62) not run.
- Stage-6 STO-13 result lost (tmpfs). Only the post-campaign STO-13b exists, on a different OS image.
- SATA bays unmeasured. Network throughput unmeasured.
- Random IO across stacks is access-method-confounded (zvol vs file). No file-based ZFS random cell exists.
- RAIDZ1 cold mixed 70/30 is PROVISIONAL (19× spread). Stage 1 cell 5 (RAIDZ2 4-wide) is incomplete.
- Stage 7: the attended rebuild arm with per-member io_ticks, which the control decision asked for, has not run. Only two control replicates exist.
- Vendor floor conditions unknown. Vulkan vs HIP gaps (#46/#48) are characterised, not explained.
- The btrfs raid10 low cold read (3.616 GB/s) is unexplained.
- NM790 endurance (TBW consumed by the campaign) is not sealed.

Each of these is tracked as a follow-up in the lab’s private issue tracker.

10 · What changed since the 2026-09-18 review

Topic	2026-09-18	Now (sealed)
mdadm raid5/6 cold seq read	1.14 GB/s, HOLD (“8× slower”)	9.035 GB/s after Gen4 reruns; the cause was SN1 at Gen1 (#72)
Fastest measured	btrfs raid5 (sole frontier)	Tie: btrfs raid5 9.034 / mdadm raid5 9.035 GB/s
Degraded / rebuild	no data	8 arms + pilot + 2 controls; RAIDZ1 93.51 % degraded read vs btrfs raid5 5.87 %
ZFS vs btrfs write	btrfs raid5 writes ~5.7× RAIDZ1	Confounded by drive state: in matched STO-09 S1, RAIDZ1 4.499 vs btrfs 2.78 GB/s
AI coverage	4 provisional cells	21 passed cells: HIP + Vulkan, streamed models, vendor-floor ratios, depth to 55k (131k via Gufo)
USB	missing	STO-13 lost; STO-13b post-campaign 905.6 / 376.2 MB/s
Write hole · SLOG	caveat · untested	unchanged: still not run

A · Evidence tables

Run ids are timestamps of sealed result directories (appendix tables omit the 2026 year). Red = non-steady; grey = no steadiness verdict; △ = PROVISIONAL. Full sealed precision is kept in the lab's private metrics ledger.

A1 · Cold regime, published Protocol B cases

layout	members	TiB	seq R	seq R	seq W	seq W	r4K R	r4K W	r16K R	r16K W	mixed	sync	smallfiles	runs
			j1	j4	j1	j4								
			GB/s	GB/s	GB/s	GB/s	IOPS	IOPS	IOPS	IOPS	IOPS	IOPS	GB/s	
ZFS RAID10 · 4×x1	4×x1	6.71	6.219	7.024	0.638	0.894	115,855	40,740	51,676	51,195	104,485	1,675	0.521	09-13-15.49
ZFS RAIDZ1 · 5-wide	4×x1 + 1×x4	14.11	6.748	7.83	0.425	0.718	67,044	18,865	19,666	30,495	757 △	415	0.534	09-13-17.14
ZFS RAIDZ2 · 5-wide	4×x1 + 1×x4	10.21	8.053	8.911	0.464	0.961	81,540	10,357	26,937	10,912	27,755	428	0.532	09-13-18.42 09-13-21.52
btrfs raid10 · 4×x1	4×x1	6.99	3.601	3.616	1.95	2.372	555,448	91,074	153,304	35,553	88,085	2,153	0.519	09-14-05.18 09-26-00.17
btrfs raid5 · 5-wide	4×x1 + 1×x4	14.43	9.033	9.034	1.912	4.066	573,194	55,450	189,731	22,981	48,472	1,365	0.44	09-14-13.57
btrfs raid6 · 5-wide	4×x1 + 1×x4	10.71	9.025	9.026	1.465	2.72	573,937	43,316	215,280	22,066	44,116	1,183	0.418	09-14-17.56
mdadm raid10 · 4×x1	4×x1	7.02	7.223	7.223	2.483	2.521	575,998	33,934	354,299	26,508	78,582	323	0.724	09-25-16.03
mdadm raid5 · 5-wide	4×x1 + 1×x4	14.41	9.034	9.035	4.411	5.04	530,451	72,598	89,454	2,489	106,552	934	0.71	09-25-18.53
mdadm raid6 · 5-wide	4×x1 + 1×x4	10.71	9.027	9.029	3.415	3.451	514,941	25,541	268,724	2,313	17,213	2,256	0.469	09-25-21.38
ZFS RAID0 · control	4×x1 + 1×x4	17.85	7.95	8.695	0.599	1.897	121,060	54,612	47,480	59,950	107,686	2,267	0.541	09-13-14.28
single SN1 · x1	SN1 @x1	3.0	1.782	1.8	0.615	0.641	35,116	14,114	13,875	16,762	22,478	1,283	0.461	09-13-13.00
single SN5 · x4	SN5 @x4	3.0	5.665	5.888	0.702	1.328	41,318	20,551	18,343	47,112	29,634	1,398	0.514	09-12-19.42
ZFS RAID10 + spare	4×x1	6.71	6.232	6.629	0.757	1.172	111,501	25,048	44,982	38,956	83,997	1,631	0.54	09-13-04.48
ZFS mirror 2-way	2×x1	3.19	3.377	3.455	0.635	0.747	71,094	17,632	32,099	20,693	51,891	1,094	0.557	09-12-05.05 09-12-17.37 09-12-22.05

A2 · ZFS primarycache=metadata

layout	members	TiB	seq R	seq R	seq W	seq W	r4K R	r4K W	r16K R	r16K W	mixed	sync	smallfiles	runs
			j1	j4	j1	j4								
			GB/s	GB/s	GB/s	GB/s	IOPS	IOPS	IOPS	IOPS	IOPS	IOPS	GB/s	
ZFS RAID10	4×x1	6.71	7.133	7.221	0.425	0.879	183,883	34,483	191,195	34,180	71,876	1,792	0.52	09-12-09.00

layout	members	TiB	seq R	seq R	seq W	seq W	r4K R	r4K W	r16K R	r16K W	mixed	sync	smallfiles	runs
			j1	j4	j1	j4								
			GB/s	GB/s	GB/s	GB/s	IOPS	IOPS	IOPS	IOPS	IOPS	IOPS	GB/s	
ZFS RAIDZ1 5w	4×x1 + 1×x4	14.11	7.252	8.215	0.483	0.664	100,690	26,464	78,651	32,929 ^Δ	35,254	256	0.558	09-13-08.49 09-13-22.49
ZFS RAIDZ2 5w	4×x1 + 1×x4	10.21	9.022	9.028	0.366	0.943	122,097	25,213	83,469	19,022	18,157	368	0.562	09-13-10.13 09-13-22.26
ZFS RAID0	4×x1 + 1×x4	17.85	9.013	8.979	0.618	2.002	185,497	52,320	168,814	67,998	102,521	2,243	0.576	09-13-07.30
ZFS RAID10 + spare	4×x1	6.71	7.134	7.222	0.854	1.14	166,324	32,942	173,687	29,288	69,147	1,699	0.514	09-13-11.37
ZFS mirror 2-way	2×x1	3.01	3.611	3.611	0.511	0.552	127,647	16,285	127,455	14,916	32,438	1,250	0.554	09-12-12.50
single x1	SN1 @x1	3.0	1.808	1.808	0.705	0.934	62,656	22,869	63,139	22,276	35,945	1,279	0.572	09-12-06.23
single x4	SN5 @x4	3.0	6.149	6.201	1.201	1.44	68,879	23,649	68,522	40,735	41,130	1,337	0.552	09-13-06.11

A3 · Warm FS controls (DRAM-ceiling regime, not layout evidence)

layout	members	TiB	seq R	seq R	seq W	seq W	r4K R	r4K W	r16K R	r16K W	mixed	sync	smallfiles	runs
			j1	j4	j1	j4								
			GB/s	GB/s	GB/s	GB/s	IOPS	IOPS	IOPS	IOPS	IOPS	IOPS	GB/s	
btrfs raid10	4×x1	7.42	30.606	35.685	2.381	12.109	4,371,901	163,146	2,416,415	156,718	537,308	3,507	0.588	09-14-12.44
btrfs raid5	4×x1 + 1×x4	14.87	29.893	36.272	2.984	3.265	4,319,328	74,417	2,420,503	86,081	194,837	3,112	0.494	09-14-16.40
btrfs raid6	4×x1 + 1×x4	11.15	29.682	35.706	2.619	2.767	4,362,546	8,744	2,417,784	9,310	28,002	1,607	0.19	09-14-19.15
mdadm raid10	4×x1	7.37	30.266	36.74	11.703	10.923	4,350,998	66,697	2,396,829	63,234	201,017	325	0.785	09-25-17.29
mdadm raid5	4×x1 + 1×x4	14.76	30.789	35.941	11.257	10.93	4,392,225	99,321	2,403,942	79,364	250,191	664	0.66	09-25-20.14
mdadm raid6	4×x1 + 1×x4	11.07	31.36	36.714	12.641	10.12	4,311,695	145,638	2,413,874	137,587	473,307	800	0.697	09-25-23.00

Stage 7 STO-09 arms (one member failed and replaced; 50 % fill). Ratios exactly as sealed in each arm’s degraded block.

arm	victim	rebuild s	rebuild GB/s	S1 seq R GB/s	S2 seq R %	S4 seq R %	S1 seq W GB/s	S2 seq W %	S4 seq W %	verdict
ZFS RAIDZ1	SN2	1740.0	1.1656	7.434	93.51	102.81	4.499	64.28	39.32	FAIL
ZFS RAIDZ1 · loaded	SN2	1550.0	1.3828	7.129	97.11	102.88	4.43	73.88	47.04	FAIL
ZFS RAIDZ2	SN2	1690.0	1.186	8.865	59.55	101.76	3.133	78.65	46.95	FAIL
btrfs raid10	SN2	2790.0	0.7316	4.432	71.19	112.7	2.316	91.41	92.75	PASS
btrfs raid5	SN2	3720.0	0.5498	9.045	5.87	95.26	2.78	98.03	109.28	PASS
btrfs raid5 · loaded	SN2	3480.0	0.6163	9.044	5.87	93.17	2.657	100.26	86.04	FAIL
btrfs raid5 · x4 victim	SN5	2590.0	0.7891	9.045	5.86	94.78	2.863	106.53	75.45	FAIL

arm	victim	rebuild s	rebuild GB/s	S1 seq R GB/s	S2 seq R %	S4 seq R %	S1 seq W GB/s	S2 seq W %	S4 seq W %	verdict
btrfs raid6	SN2	6380.0	0.3207	9.042	5.86	96.0	1.968	117.94	32.67	FAIL
no-failure control · run 2026-09-25-11.29	–	–	–	7.622	97.55	99.51	4.24	66.67	66.58	S4/S2 0.9986
no-failure control · run 2026-09-26-00.58	–	–	–	7.409	103.14	98.65	4.397	66.59	67.71	S4/S2 1.0167

Ours (vendor-invocation group, cold btrfs raid5) vs “Minisforum reports” floor (vendor product page, via the lab’s vendor-floor register). Ratio = ours ÷ floor, as sealed.

model	backend	ours pp512	floor pp512	ratio	ours tg128	floor tg128	ratio	run
gpt-oss-20b	HIP	1,514.00	1575.75	0.9608	75.31	83.09	0.9064	2026-09-23-13.05
gpt-oss-20b	Vulkan	1,408.84	1575.75	0.8941	80.72	83.09	0.9715	2026-09-23-07.05
Gemma 4 26B-A4B	HIP	1,148.97	1200.58	0.957	50.13	70.33	0.7128	2026-09-22-06.02
Gemma 4 26B-A4B	Vulkan	1,102.47	1200.58	0.9183	55.50	70.33	0.7892	2026-09-22-15.58
gpt-oss-120b	HIP	572.64	668.79	0.8562	53.23	59.3	0.8976	2026-09-23-15.03
gpt-oss-120b	Vulkan	551.75	668.79	0.825	57.33	59.3	0.9668	2026-09-23-08.23
Qwen3.5-122B-A10B	HIP	306.34	317.58	0.9646	23.03	31.76	0.7252	2026-09-22-10.11
Qwen3.5-122B-A10B	Vulkan	283.08	317.58	0.8914	24.47	31.76	0.7705	2026-09-23-00.06

Stage 3 AI cells: llama.cpp llama-bench card (pp512, tg128 at depth 0), cold-load rail. HIP unless “-vk”. Values exactly as sealed.

model	backend	cold load		pp512 tok/s	tg128 tok/s	pp @32k tok/s	tg @55k tok/s	run
Qwen3-0.6B Q8 (smoke)	HIP	btrfs raid5 · cold	2.025	11434.47435	229.529638	2745.061274	–	2026-09-22-05.43
Qwen3-0.6B Q8 (smoke)	HIP	ZFS RAIDZ1 · pc=all, ARC cold	3.072	11585.991494	229.846661	2777.739635	–	2026-09-23-16.11
Llama-2-7B Q4_0	HIP	btrfs raid5 · cold	2.025	1366.272992	54.158066	432.078239	–	2026-09-29-01.31
gpt-oss-20b MXFP4	HIP	btrfs raid5 · cold	5.04	1488.023566	75.491938	1071.686866	45.28471	2026-09-23-13.05
gpt-oss-20b MXFP4	Vulkan	btrfs raid5 · cold	6.761	1408.366322	80.697347	790.282376	53.880727	2026-09-23-07.05
Gemma 4 26B-A4B Q4_K_M	HIP	btrfs raid5 · cold	5.035	1130.874349	50.146229	733.241665	37.31528	2026-09-22-06.02
Gemma 4 26B-A4B Q4_K_M	HIP	ZFS RAIDZ1 · pc=all, ARC cold	6.758	1136.82361	50.079649	721.198722	37.280024	2026-09-23-04.57
Gemma 4 26B-A4B Q4_K_M	Vulkan	btrfs raid5 · cold	10.654	1093.213531	55.40021	730.436897	39.273202	2026-09-22-15.58
Qwen3.8-27B Q4_K_XL	HIP	btrfs raid5 · cold	4.635	332.939424	12.589999	278.517628	10.553017	2026-09-29-18.07
Qwen3.8-27B Q8_K_XL	HIP	btrfs raid5 · cold	6.782	–	–	–	–	2026-09-29-09.55
gpt-oss-120b MXFP4	HIP	btrfs raid5 · cold	11.256	564.012211	53.116634	457.81774	31.240673	2026-09-23-15.03
gpt-oss-120b MXFP4	Vulkan	btrfs raid5 · cold	49.375	548.470986	57.253699	373.161288	37.666248	2026-09-23-08.23
Qwen3.5-122B-A10B Q4_K_M	HIP	btrfs raid5 · cold	12.111	308.208797	22.969954	264.913976	19.943198	2026-09-22-10.11
Qwen3.5-122B-A10B Q4_K_M	Vulkan	btrfs raid5 · cold	63.25	279.392725	24.455856	229.577776	21.093569	2026-09-23-00.06

model	backend	pool	cold load s	pp512 tok/s	tg128 tok/s	pp @32k tok/s	tg @55k tok/s	run
DeepSeek V4 Flash IQ2_XXS	HIP	btrfs raid5 · cold	13.65	27.353744	15.261516	–	–	2026-09- 29-16.41
DeepSeek V3.1 Q2_K_XL (streamed)	HIP	btrfs raid5 · cold	5.122	14.67113796370338	1.782020828368677	–	–	2026-09- 29-15.17
DeepSeek V3.1 Q2_K_XL (streamed)	HIP	ZFS RAIDZ1 · pc=all, ARC cold	6.556	13.500959553793148	1.4079943045119931	–	–	2026-09- 29-05.00
Kimi K2 Thinking IQ1 (streamed)	HIP	btrfs raid5 · cold	38.334	3.569684	1.093135	–	–	2026-09- 29-03.21
Kimi K2 Thinking IQ1 (streamed)	HIP	ZFS RAIDZ1 · pc=all, ARC cold	181.147	4.688261	1.548387	–	–	2026-09- 29-13.35

Steadiness heat strip

Each cell = one published case. Teal = rails.steadiness.steady true; coral = false (reported, labelled); grey = no verdict; blank = not run.

	seqR j1	seqR j4	seqW j1	seqW j4	r4K R	r4K W	r16K R	r16K W	r64K R	mixed	sync	smallf.	VM mix
ZFS RAID10													
ZFS RAIDZ1										Δ			
ZFS RAIDZ2													
btrfs raid10													
btrfs raid5													
btrfs raid6													
mdadm raid10													
mdadm raid5													
mdadm raid6													
ZFS RAID0													
single SN1													
single SN5													
ZFS RAID10 · meta													
ZFS RAIDZ1 · meta								Δ	Δ				
ZFS RAIDZ2 · meta													
btrfs raid5 · warm													
btrfs raid6 · warm													
mdadm raid5 · warm													
steady 182 · non-steady 25 · no verdict 18													

STO-06 smallfiles has no steadiness verdict by design (single-stream copy). Warm rows converge near the DRAM ceiling and are not layout evidence.

Steadiness of every published case used in the appendix tables.

Built 2026-10-08 22:11 PT from the lab’s sealed metrics ledger (2519 rows). Drive labels SN1–SN5 only.